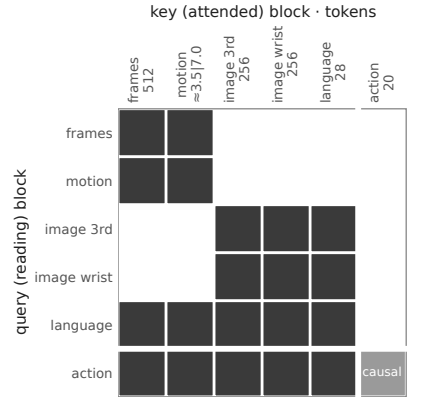
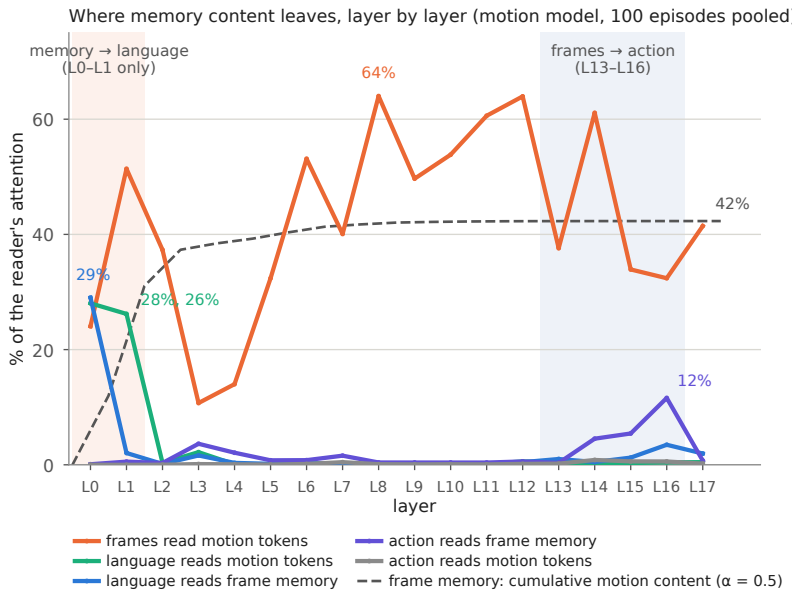
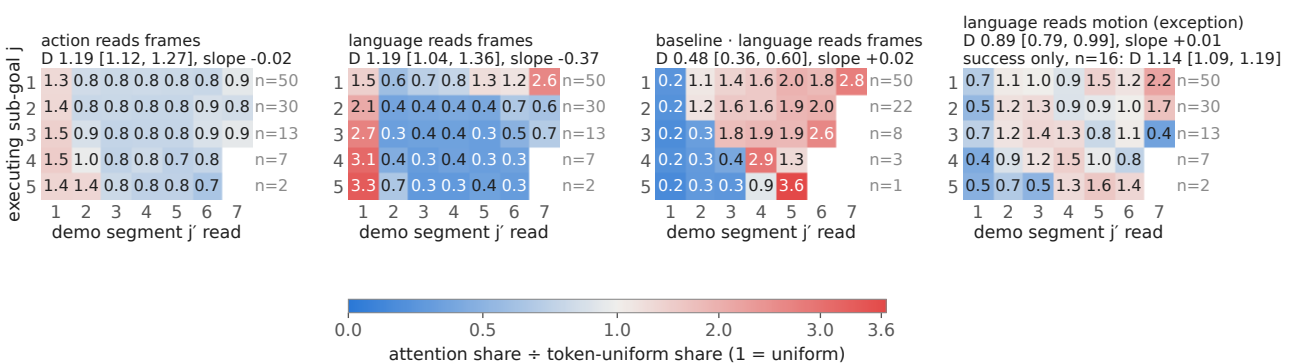
I. Two attention pathways carry motion-token content into the action tokens

Motion model, PatternLock | RouteStick, seed 42, 100 episodes / 495 inferences, head-averaged, episode-equal. (a) the attention mask as a block matrix; (b) per layer, how much language, action and frame memory read from motion tokens and frame memory (both tasks pooled).



Block attention mask (dark = may attend; verified as exactly 0 elsewhere in every layer, head, episode). prefix = memory → images → language; suffix = action.



II. The pathways exist, but which demo segment is read shows no sub-goal alignment

RouteStick, motion model unless noted, 50 episodes per model, all inferences; rows pool episodes with different numbers of sub-goals m (n = episodes per row); frame stream weighted 'single', motion stream 'overlap'; D = diagonal enrichment, episode-level 95% CI.